

# Lab 3: ACF / PACF

Reference solution

Aymen Ben Brik  
aymen.benbrik@esprit.tn  
Esprit School of Business

2025–2026

Reference Python code in *correction.py*.

## Exercise 1 – White noise

### Solution

On 500 i.i.d.  $\mathcal{N}(0, 1)$  draws (seed 42), the empirical ACF spikes at  $|\hat{\rho}(h)|$  are essentially all within  $1.96/\sqrt{500} \approx 0.088$ . Out of 30 lags tested, typically 1 or 2 exceed the band — close to the expected  $0.05 \times 30 = 1.5$  false positives.

## Exercise 2 – AR(1) simulation

### Solution

Empirical ACF agrees well with  $\rho(h) = \phi^h$  for  $\phi = 0.3, 0.7, -0.7$ . For  $\phi = 0.95$ , the empirical ACF *decays slower* than the theoretical exponential due to finite-sample bias: with  $T = 500$  the series is close to a random walk, and  $\hat{\rho}$  often overestimates persistence at high lags.

The empirical PACF for  $\phi = 0.7$  shows  $\hat{\alpha}(1) \approx 0.7$  and  $\hat{\alpha}(h) \approx 0$  for  $h \geq 2$  — the cutoff signature of an AR(1).

## Exercise 3 – MA(1) simulation

### Solution

Theoretical  $\rho(1) = \theta/(1 + \theta^2)$ :

- $\theta = 0.5$ :  $\rho(1) = 0.5/1.25 = 0.4$ .
- $\theta = -0.5$ :  $\rho(1) = -0.4$ .
- $\theta = 0.9$ :  $\rho(1) = 0.9/1.81 \approx 0.497$ .

Empirical ACF on  $T = 500$  samples confirms these values to within  $\pm 0.05$ . PACF decays geometrically, never cuts off — opposite signature to AR(1).

## Exercise 4 – Atlanta ACF/PACF

**Solution**

(1) The raw ACF of the Atlanta series shows strong, slowly decaying autocorrelations at lags 12, 24, 36, ... (annual seasonality) and high values at small lags from the remaining trend. The series is far from white noise.

(2) After decomposition (centred MA  $k = 6$  + seasonal averaging) the residual standard deviation is  $\approx 2.88^\circ\text{F}$ .

(3) The residual ACF is much cleaner: the seasonal spikes disappear, but a few short-lag autocorrelations remain (the residuals are approximately AR(1) with  $\phi \approx 0.3$ ). The Atlanta residual is *not exactly white noise* — there is short-term weather persistence to capture with an ARMA model (Chapter 5).

**Exercise 5 – Ljung–Box****Solution**

On the Atlanta residual, the Ljung–Box statistic at  $m = 12$  is  $\approx 200$  with  $p$ -value  $\approx 0$ : **reject** the white-noise hypothesis. The same conclusion holds at  $m = 24, 36$ .

On the simulated white noise (Exercise 1),  $Q_{\text{LB}}(m = 12) \approx 6\text{--}15$  with  $p$ -value  $> 0.10$ : **do not reject** — correct behaviour under  $H_0$ .